

Objective:

To demonstrate IllegalStateException

Execution Steps:

STEP 1:

Open IntelliJ IDEA.

STEP 2:

Create a Java project.

STEP 3:

Create a package named:

javacore

STEP 4:

Create a Java class named:

IllegalStateException

STEP 5:

Write the given program inside the class.

STEP 6:

Save the program.

STEP 7:

Compile the program.

STEP 8:

Run the program.

STEP 9:

Execution starts from the `main()` method.

STEP 10:

A Scanner object named `sc` is created.

STEP 11:

The try block starts executing.

STEP 12:

`sc.close()` closes the Scanner object.

STEP 13:

The program tries to execute `sc.nextInt()`.

STEP 14:

The Scanner is already closed, so it is in an invalid state.

STEP 15:

`IllegalStateException` occurs.

STEP 16:

The catch block catches the exception.

STEP 17:

"Illegal State Exception" is displayed.

STEP 18:

"Program ended" is displayed.

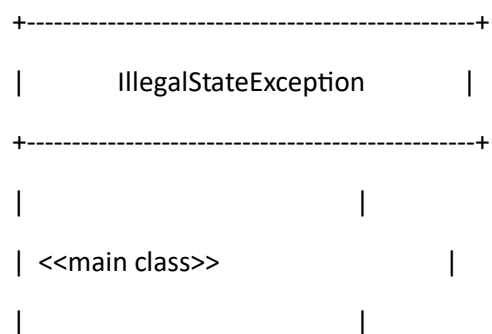
STEP 19:

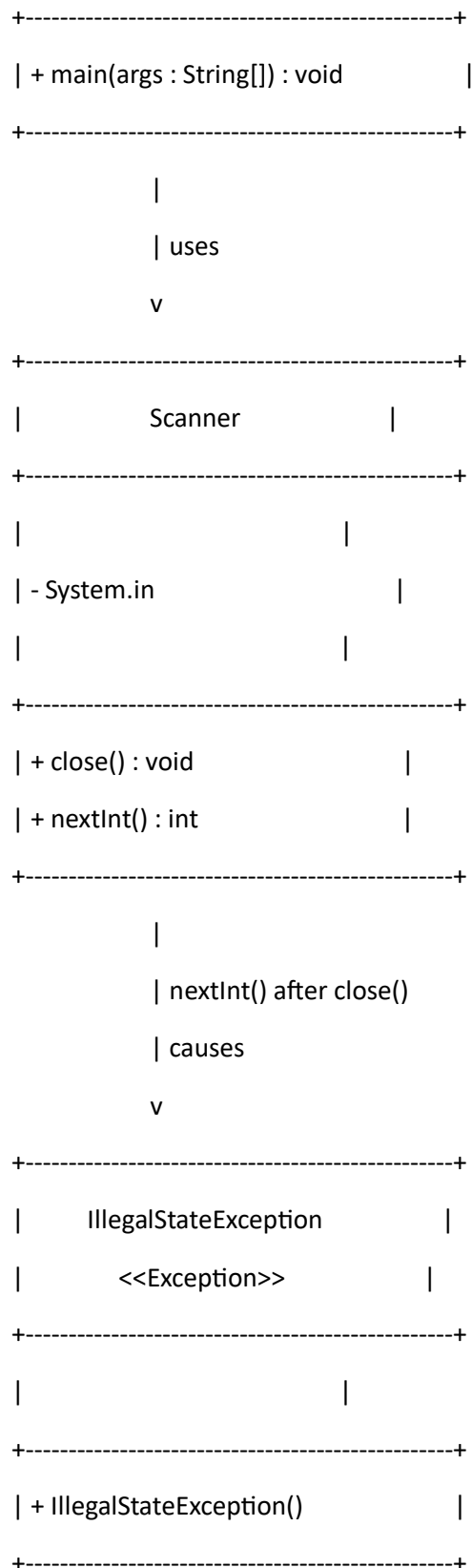
The program ends.

Algorithm:

1. Start the program.
2. Create a Scanner object.
3. Start the try block.
4. Close the Scanner object using close().
5. Try to read an integer using nextInt().
6. Since the Scanner is already closed, `IllegalStateException` occurs.
7. Catch the exception.
8. Display "Illegal State Exception".
9. Display "Program ended".
10. Stop the program.

UML Class Diagram:





Code:

```
package javacore;
```

```
import java.util.Scanner;

public class IllegalStateException {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            sc.close();

            System.out.println(sc.nextInt());
        }
        catch (IllegalStateException e) {
            System.out.println("Illegal State Exception");
        }

        System.out.println("Program ended");
    }
}
```

Output:

Illegal State Exception

Program ended